Environmental Risk Assessment

Dashboard pages, 2015 to 2023

This document shows the four report pages of the Power BI file, with a guide to what each one shows and how to read it. The data is 500 readings from seven marine locations, covering 1 January 2015 to 31 December 2023.

For the method, the bleaching check and the limitations, read Environmental_Risk_Assessment.pdf.

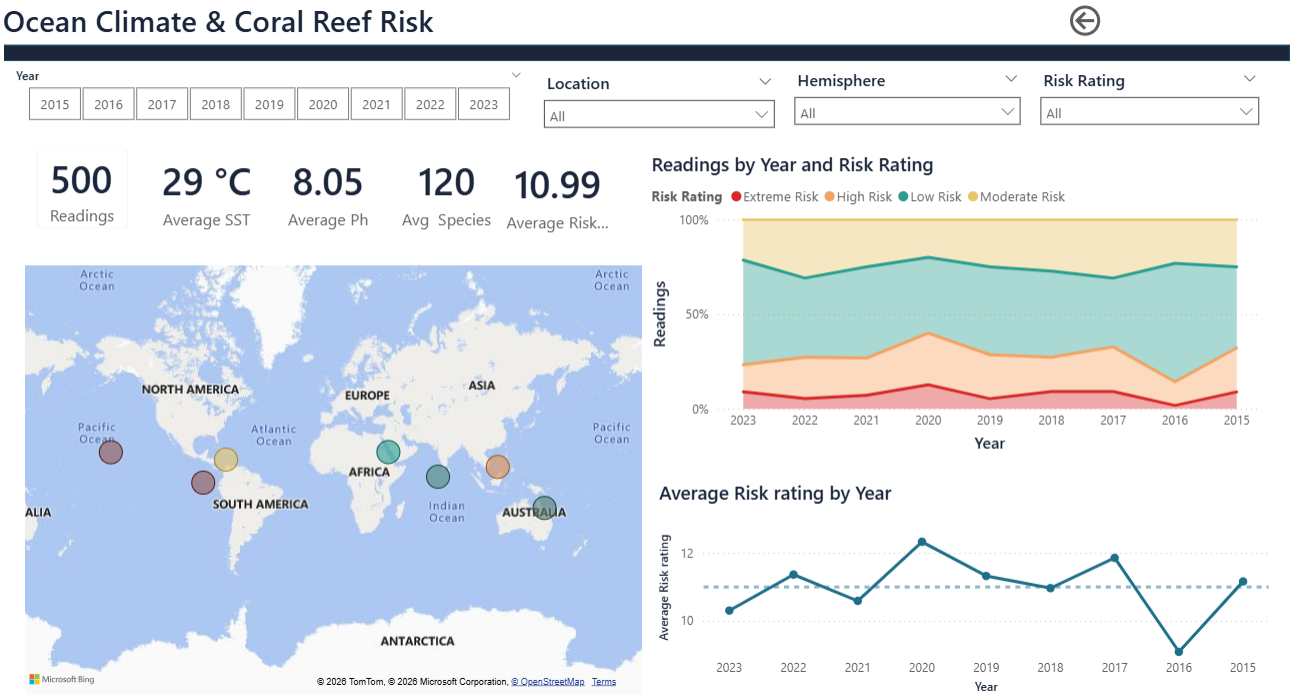
Pages in this document

- Overview
- Drivers
- 2020 vs 2021
- Location Deep Dive (drill-through)

Note on the screenshots

The screenshots were taken from Power BI Desktop before the final edits. Some still show the earlier title, 'Ocean Climate & Coral Reef Risk', and a few labels, number formats and sort orders may differ from the finished report. Re-export the pages after the final edits to update them.

1. Overview



What it shows

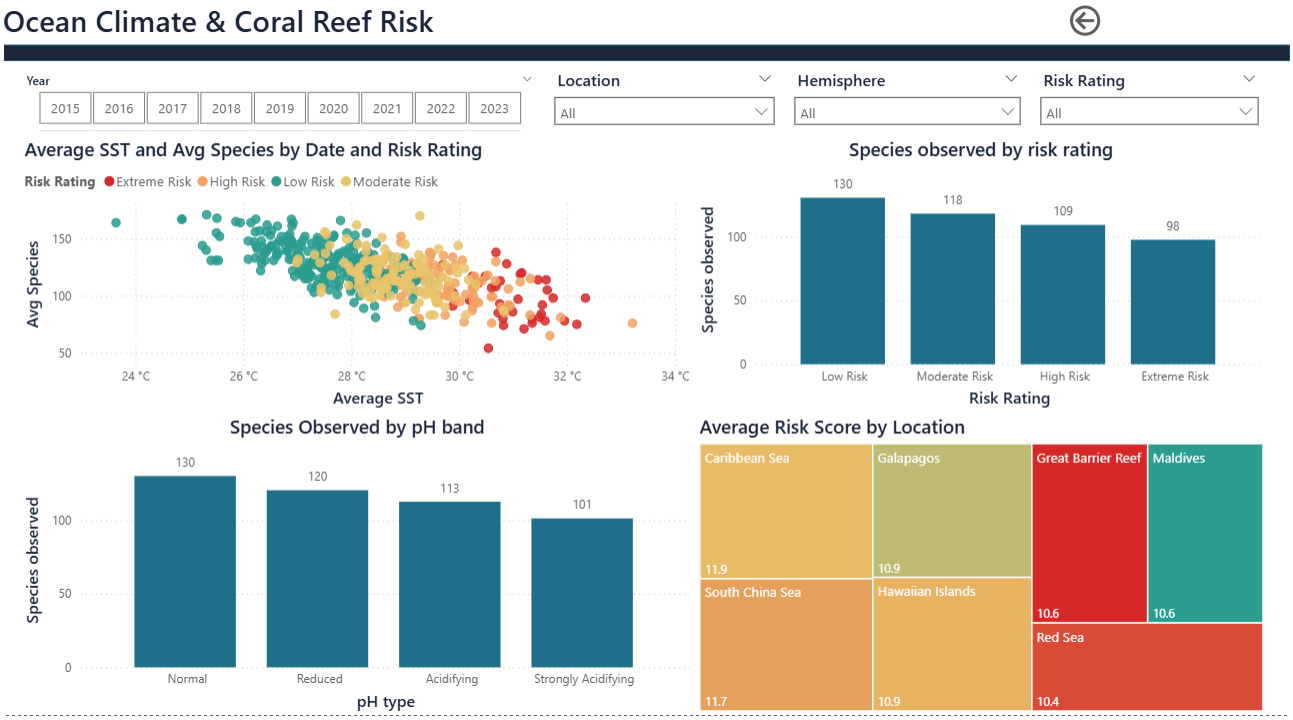
- Four cards for all 500 readings: average SST 29 °C, average pH 8.05, average species 120, and average risk score 10.99.
- Readings by year and risk rating, as a 100% area chart. The High and Extreme share peaks in 2020 at 40%, then drops to 27% in 2021.
- Average risk by year, with the period average as a dashed line. The highest year is 2020 (12.3) and the lowest is 2016 (9.1).
- A map of the seven locations.

How to read it

- Use the Year, Location, Hemisphere and Risk Rating slicers to filter every visual on the page.
- After the 2020 peak, risk falls below the average in 2021 (10.6) and 2023 (10.3). It rises again in 2022 (11.4).

Source: Power BI Desktop, page 'Overview'.

2. Drivers



What it shows

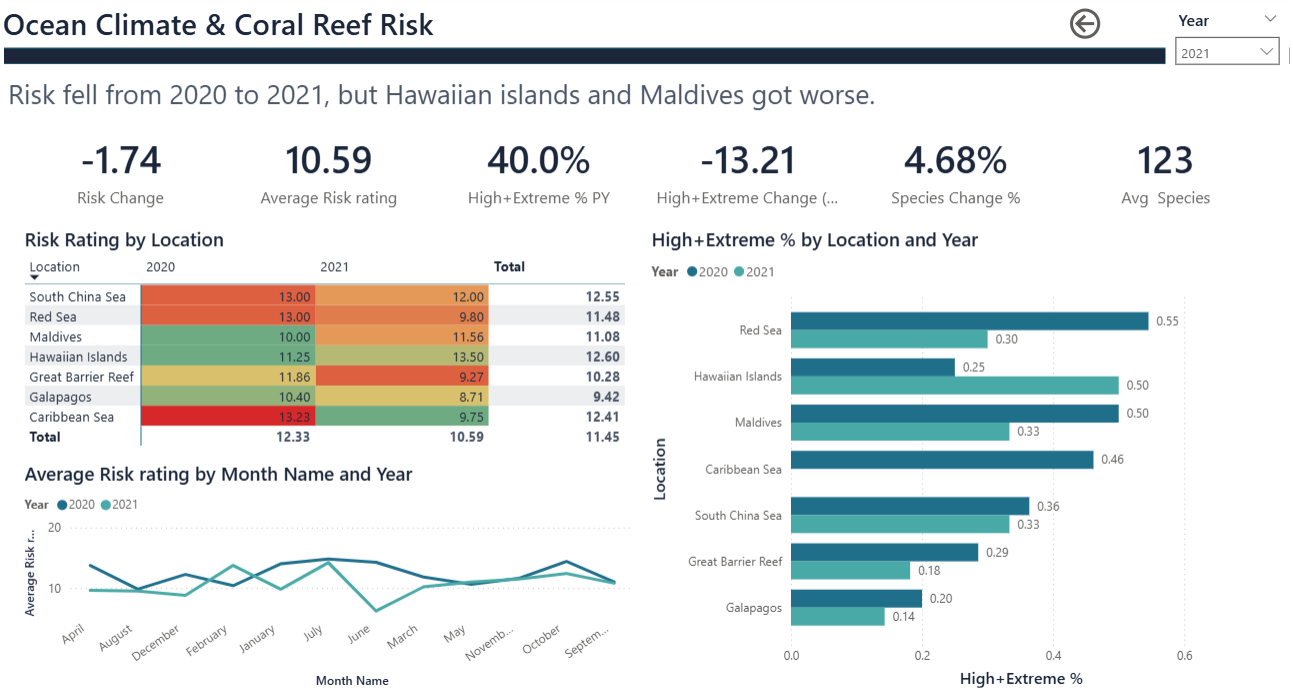
- Average SST against average species, one point per reading, coloured by risk rating. Species observed fall as SST rises.
- Species observed by risk rating: 130 at Low, 118 at Moderate, 109 at High and 98 at Extreme.
- Species observed by pH band: 130 at Normal, 120 at Reduced, 113 at Acidifying and 101 at Strongly acidifying.
- Average risk score by location, as a treemap. Caribbean Sea is highest (11.9) and Red Sea is lowest (10.4).

How to read it

- Species fall at each step up in risk rating and at each step down in pH. Both drivers point the same way. The risk score is built from SST and pH, so part of this link is built in.

Source: Power BI Desktop, page 'Drivers'.

3. 2020 vs 2021



What it shows

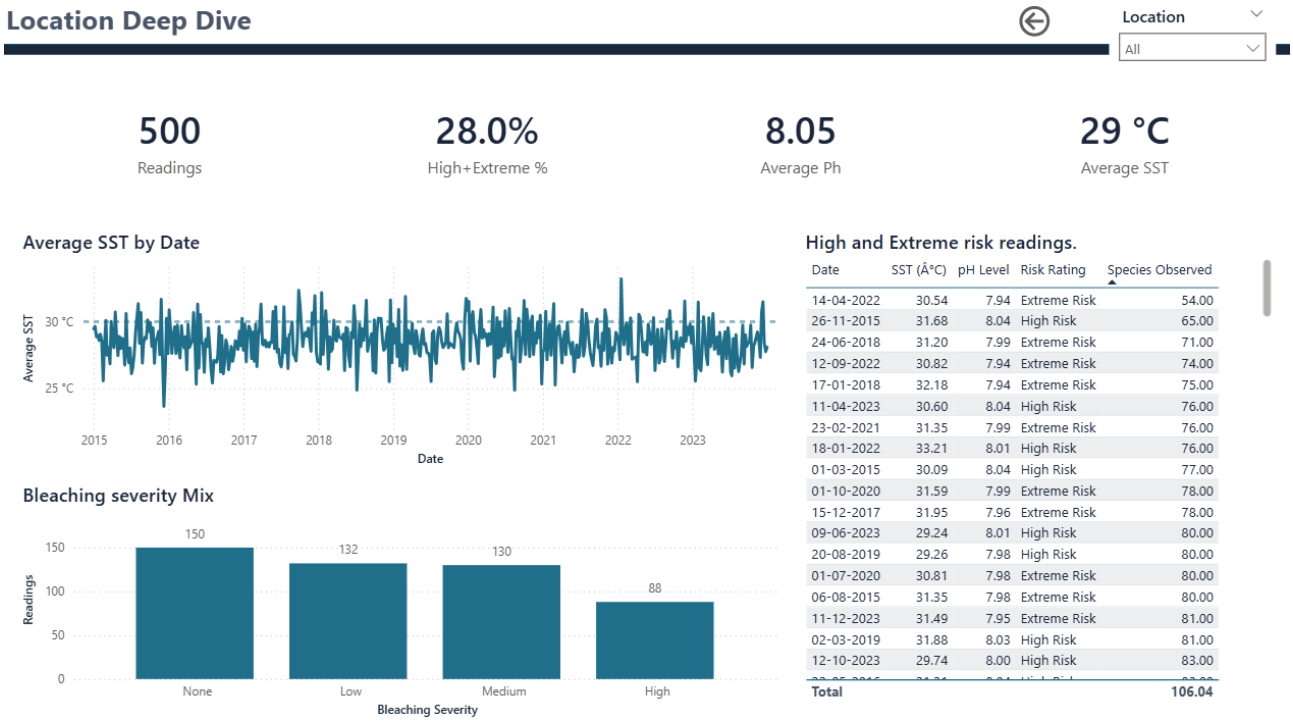
- Year cards comparing 2021 with 2020: average risk down 1.74, High or Extreme share down 13.2 points, and average species up 4.7%.
- Average risk by location and year. Hawaiian Islands (11.25 to 13.50) and Maldives (10.00 to 11.56) rose. The other five locations fell.
- High or Extreme share by location and year. Caribbean Sea has no 2021 bar because none of its four 2021 readings was High or Extreme.
- Average risk by month for 2020 and 2021.

How to read it

- One year-on-year change is not a trend. Risk rose again in 2022 (11.4), so treat the 2021 fall as a single change in a noisy series.

Source: Power BI Desktop, page '2020 vs 2021'.

4. Location Deep Dive (drill-through)



What it shows

- Cards for the selected location (All in this screenshot): 500 readings, 28.0% High or Extreme, average pH 8.05 and average SST 29 °C.
- Average SST by date, with a dashed line at 30 °C. Readings at or above the line are marine heatwave days.
- Bleaching severity counts: 150 blank, 132 Low, 130 Medium and 88 High. The report shows blanks as None. They are unrecorded, not confirmed absent.
- A table of High and Extreme readings with date, SST, pH, risk rating and species observed. The table total is the average species count.

How to read it

- Right-click a location on another page and choose Location Deep Dive from the drill-through menu. This page then opens filtered to that location.

Source: Power BI Desktop, page 'Location Deep Dive'.